

4. Implement a finite-state machine for morphological parsing. In this example, we'll create a simple machine to generate plural forms of English nouns using python.

7]  
0s

```
def pluralize(noun):  
    if noun.endswith('y'):  
        return noun[:-1] + 'ies'  
    elif noun.endswith(('s', 'x', 'z', 'ch', 'sh')):  
        return noun + 'es'  
    else:  
        return noun + 's'  
  
# Test  
words = ["cat", "bus", "baby", "box", "dish"]  
for word in words:  
    print(word, "->", pluralize(word))
```

✓

```
... cat -> cats  
    bus -> buses  
    baby -> babies  
    box -> boxes  
    dish -> dishes
```